Arpan Roy Chowdhury

Software Engineer

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Backend and platform engineer with 4+ years building APIs and services in TypeScript and Node, across payments, authentication, subscriptions, and high-traffic public APIs serving over a million users. Focused on API design that stays performant and maintainable at scale, with hands-on delivery on GCP and AWS using Docker and Kubernetes.

Experience

Mentalyc Inc Remote

Forward Deployed Engineer *Mar 2026 – Present*

- Build backend services and APIs in Node/TypeScript for an AI documentation platform — audio transcription and processing pipelines, LLM-driven generation, and REST integrations with external systems, in a HIPAA-conscious environment.
- Build RAG pipelines (LlamaIndex, LangChain, pgvector), design data models and serverless backend logic, and deploy and monitor services on cloud infrastructure.
- Improve reliability and observability of production workflows, and build internal tooling that speeds up onboarding and debugging.

Viact.ai Remote

Software Engineer *Dec 2024 – Mar 2026*

- Built backend services and APIs for an enterprise AI platform — real-time data pipelines ingesting streams from edge devices and computer-vision models, exposed through REST APIs and webhooks.
- Containerized services with Docker and deployed across cloud environments for enterprise reliability and scale.
- Tuned API performance for real-time monitoring workloads and shaped backend architecture with product and leadership for enterprise use cases.

DappRadar Remote

SDE II *Jul 2023 – Dec 2024*

- Built and maintained backend services and APIs (Node.js/PHP) aggregating and indexing on-chain data across multiple blockchains, serving dApp, token, NFT, and DeFi data to 1M+ users.
- Designed high-throughput data pipelines and performant REST APIs, optimizing for scalability, reliability, and data accuracy at scale.
- Integrated smart-contract and on-chain data sources, and hardened error handling and data freshness across multi-chain indexing.

Hakuna Remote

Application Developer *Jan 2021 – Jul 2023*

- Built backend features for a real-time live-streaming platform with 10M+ downloads — APIs for live video, chat, in-app gifting and virtual currency, and content discovery.
- Worked with WebRTC and real-time infrastructure for low-latency streaming at scale.
- Designed APIs and data models for social and monetization features, focused on reliability under high concurrency.

ET Media Labs India

Executive Systems Engineer *Mar 2020 – Nov 2020*

- Built backend systems, automation, and data pipelines for multi-channel data acquisition and analytics, integrating platform APIs.
- Automated reporting pipelines, improving reporting accuracy and turnaround for stakeholders.

Projects

Shindigs (shindigs.cc) | *Node, TypeScript, Supabase/PostgreSQL, Pinecone* Funded

- Building the backend for a funded ticketing platform — APIs for events, multiple ticket types, and pricing; phone/WhatsApp OTP auth; and a payments integration with post-purchase confirmation via the WhatsApp Business API.
- Added an AI discovery layer with Pinecone and LangChain, and a publish pipeline that keeps the public app in sync with organizer changes.

ninjatest.app | *Node, Supabase, PostgreSQL* Under development

- Building a real-time 1v1 exam-prep platform — authoritative game logic in PostgreSQL security-definer RPCs, ELO matchmaking, and synchronized state over Supabase Realtime.

Technical Skills

Languages: TypeScript, JavaScript, Node.js, Python, PHP, SQL
Backend & APIs: REST API Design, Public & Internal APIs, Auth, Payments, Subscriptions, Webhooks, OAuth, GraphQL, WebRTC
Data: PostgreSQL, Supabase, Redis, pgvector, Data Modeling, ETL, High-Throughput Pipelines
Cloud & DevOps: GCP, AWS, Docker, Kubernetes, CI/CD, Vercel, Postman
AI & Audio: LLM Orchestration, LangChain, LlamaIndex, RAG, Pinecone, Transcription/Audio Pipelines, Prompt Engineering

Education

IIT Kharagpur
Executive Post Graduate Certificate in Generative AI & Agentic AI *2025*

Maulana Abul Kalam Azad University of Technology
Bachelor of Computer Application (BCA) *Jul 2016 – Aug 2019*